

Musa's Module 2 Task Solution

Required Deliverables of the task:

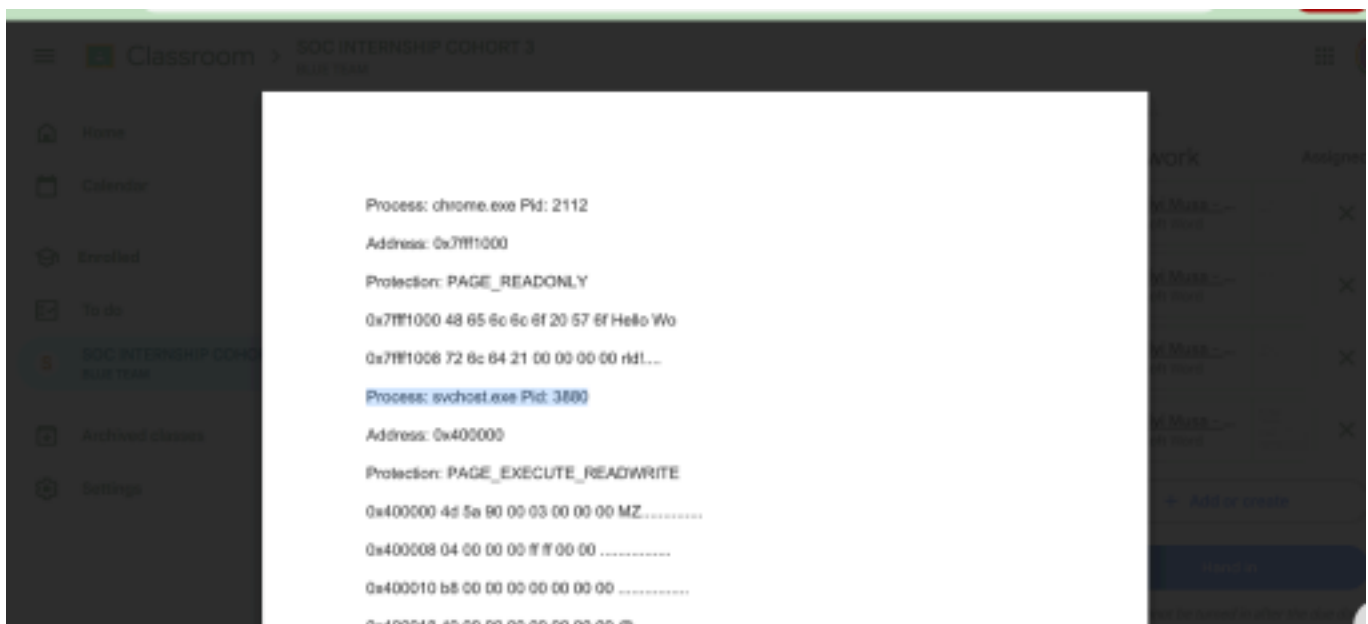
In your final investigation report, you must answer the following:

1. **Identify the Compromised Process:** Provide the Name and PID of the process acting as the vessel for the malware.
2. **Parent-Child Analysis:** Identify the Parent Process (Name and PID) of the malicious entry. Explain why this relationship is architecturally significant or suspicious in a Windows environment.
3. **Memory Evidence:** Cite the specific memory protection flags and Magic Bytes found in the malfind report that prove code injection has occurred.
4. **C2 Identification:** Provide the Destination IP address and Port the compromised process is communicating with.
5. **Attack Classification:** Based on the evidence (Process Tree + Memory Injection + Network Call), what specific type of memory-resident attack is this? (e.g., Process Hollowing, DLL Injection, or Reflective Loading).

SOLUTION

1. Identify the Compromised Process

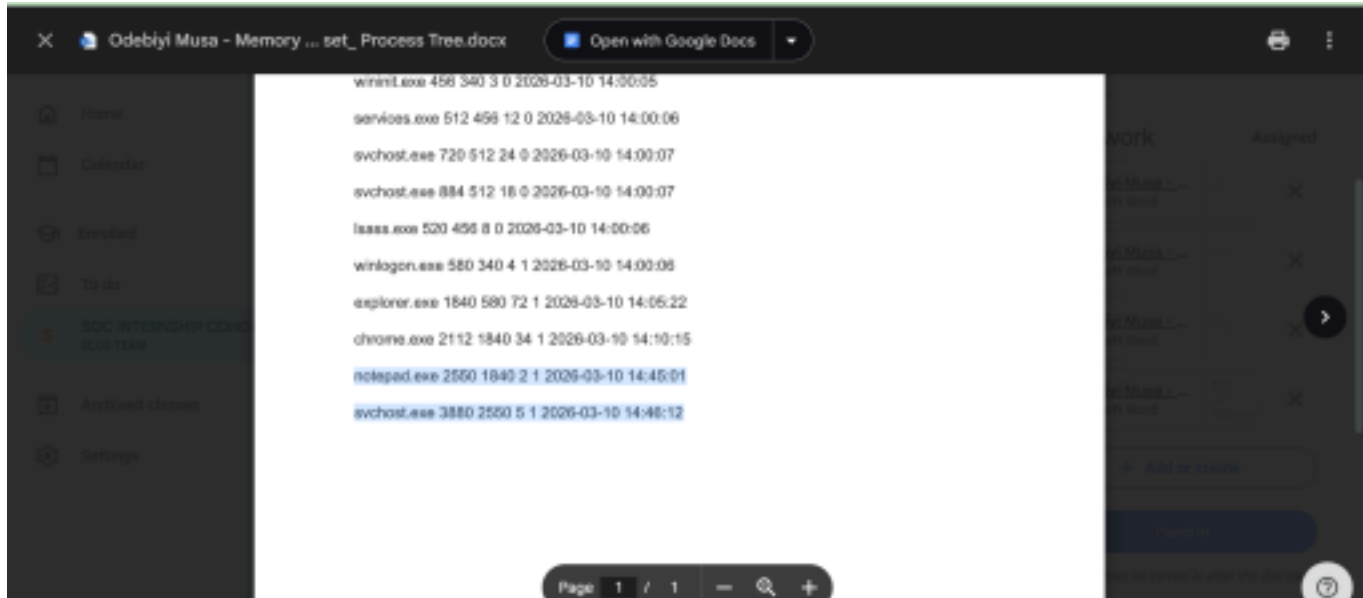
- Process Name: svchost.exe
- Process ID: 3880



2. Parent-Child Analysis

- Parent Process Name: notepad.exe
- Parent Process ID: 2550

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In a Windows environment, process creation follows a well-defined hierarchical structure where specific system processes are responsible for spawning others. For example, `svchost.exe` is typically launched by `services.exe` to host system services.

In this case, `svchost.exe` (PID 3880) is observed as a child of `notepad.exe` (PID 2550), which is architecturally abnormal. `notepad.exe` is a user-level application with no role in service management and should not spawn system service processes.

This deviation from the expected process hierarchy indicates process manipulation and is a strong indicator of malicious activity.

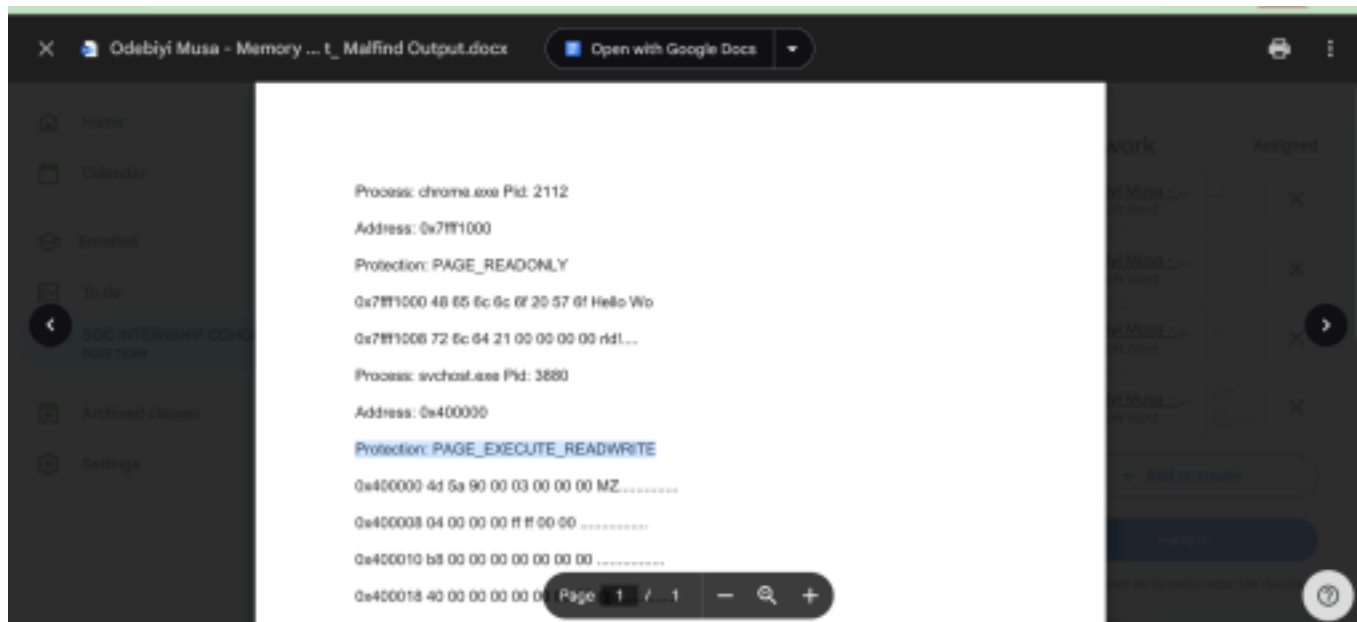
3. Memory Evidence

The memory analysis revealed two critical indicators of code injection within the process. First, the memory region was marked as `PAGE_EXECUTE_READWRITE`, indicating that it was simultaneously writable and executable, which is an abnormal configuration in Windows memory management that is commonly associated with process injection techniques.

Second, the presence of the `MZ` header confirms that a Windows Portable Executable (PE) structure exists within the process memory space. This suggests that an executable has been injected directly into memory rather than loaded from disk.

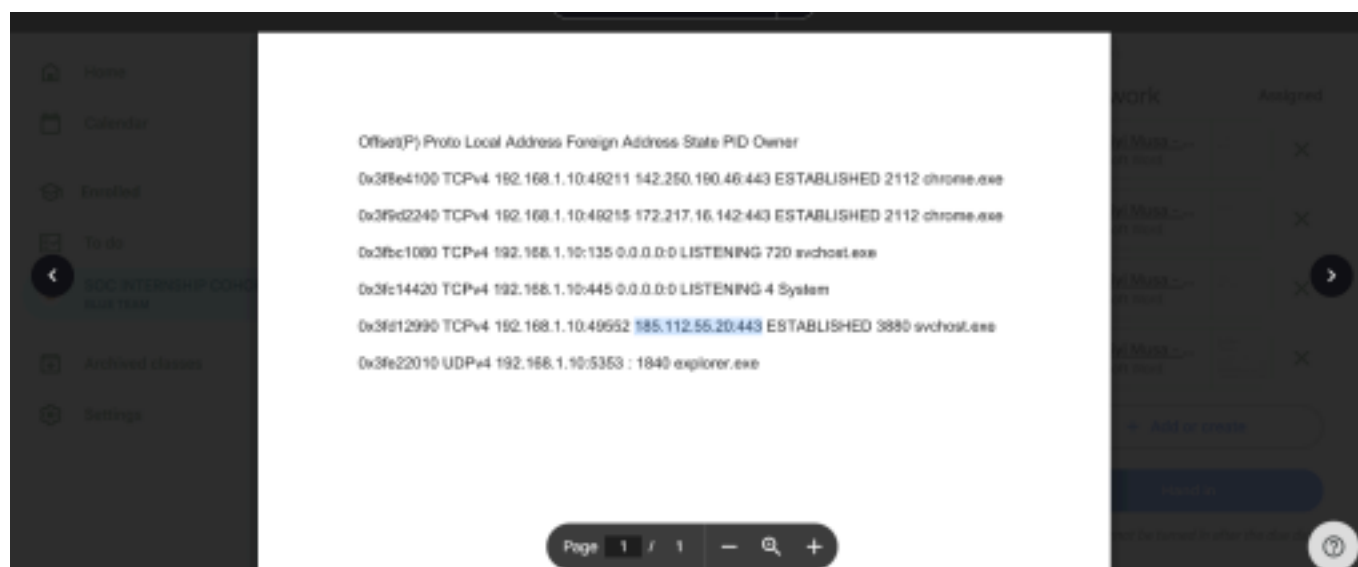
Together, these indicators provide strong evidence of in-memory code injection and fileless malware execution

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4. C2 Identification

- Destination IP address: 185.112.55.20
- Port: 443



5. Attack Classification

The observed activity is consistent with a Process Hollowing attack (MITRE ATT&CK T1055.012) where a legitimate Windows service process (**svchost.exe**) was manipulated in memory to execute malicious code. The presence of RWX memory and subsequent network communication confirms a fileless, memory-resident attack chain with likely Command-and-Control activity (T1071.001).

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